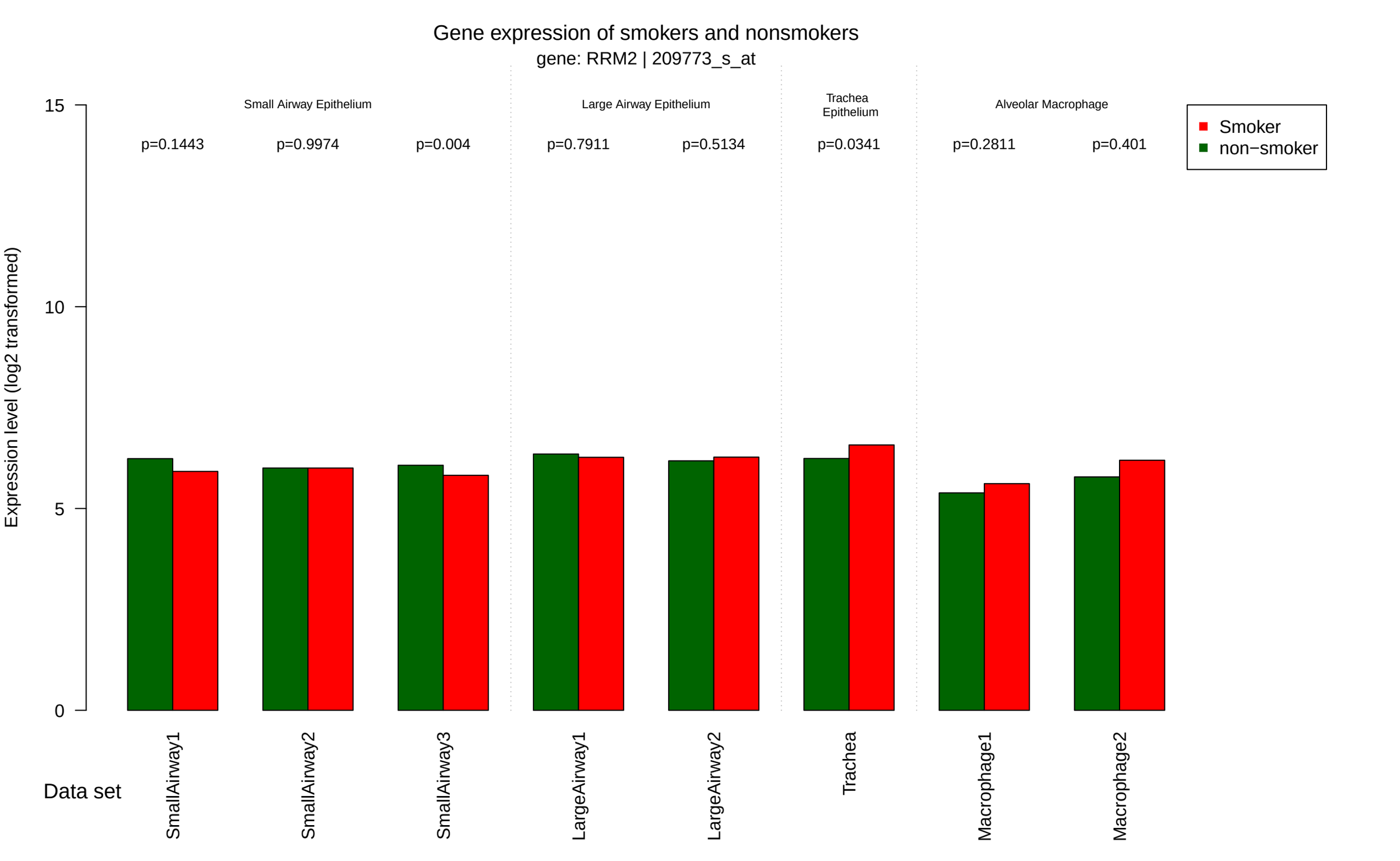
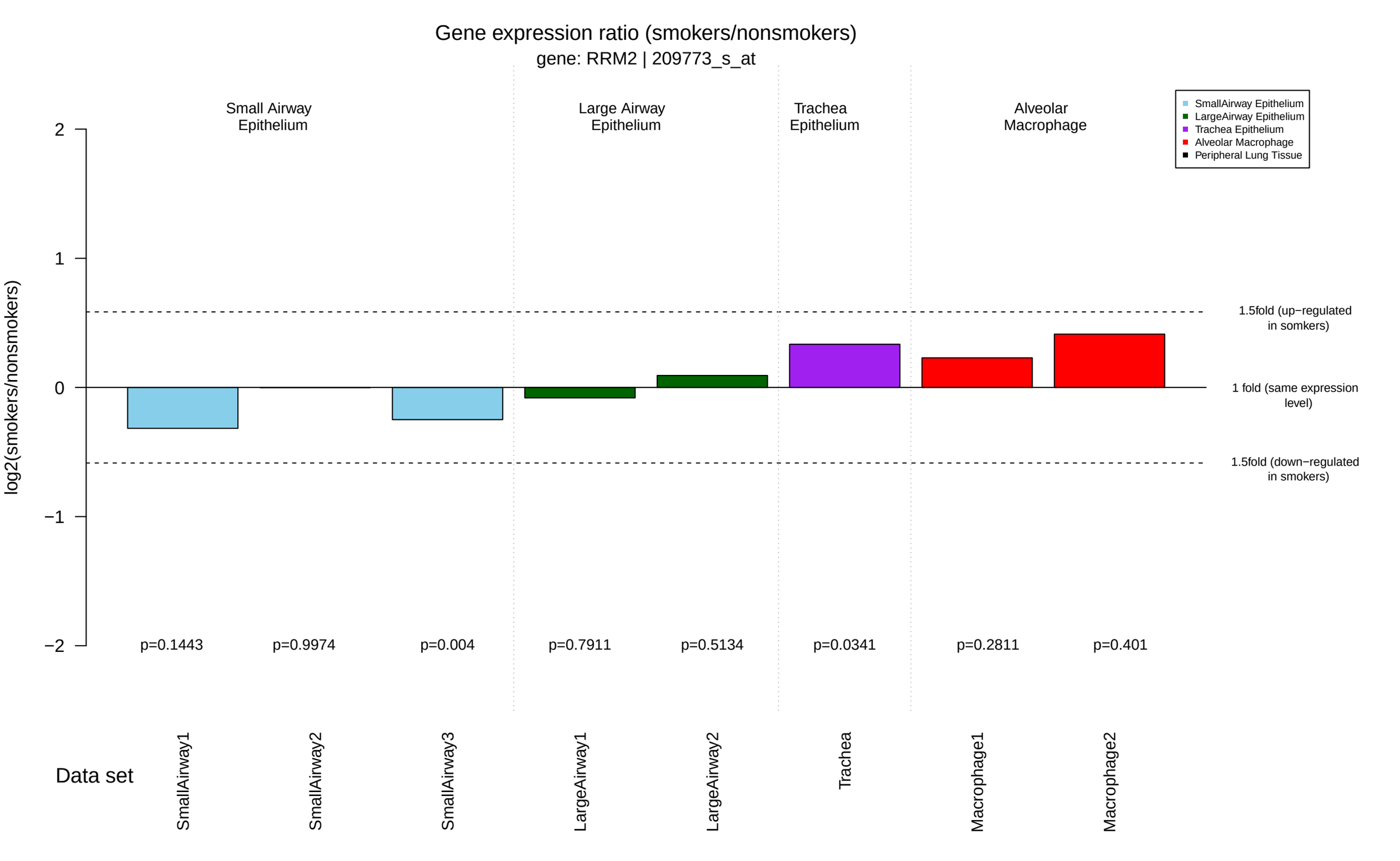
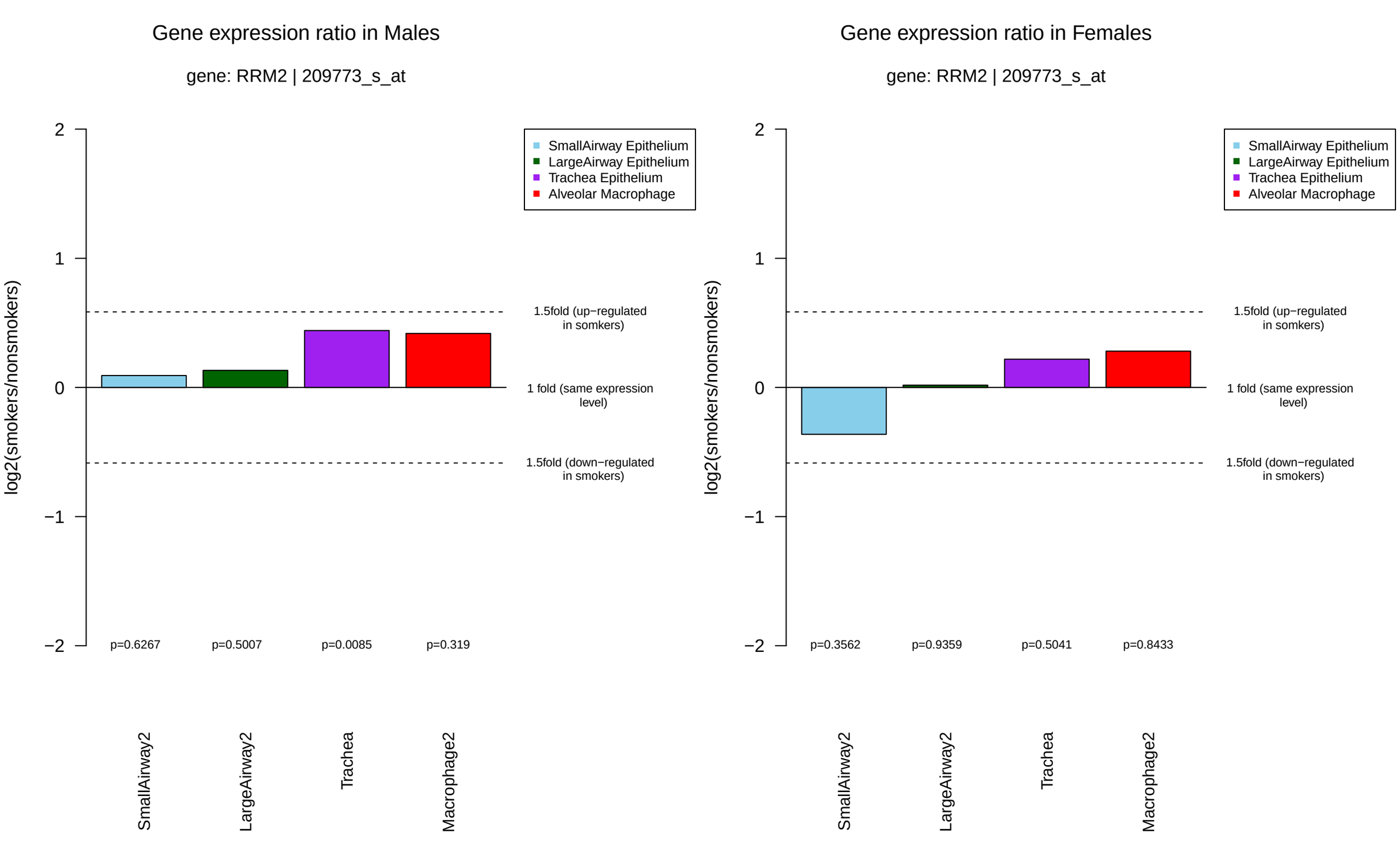
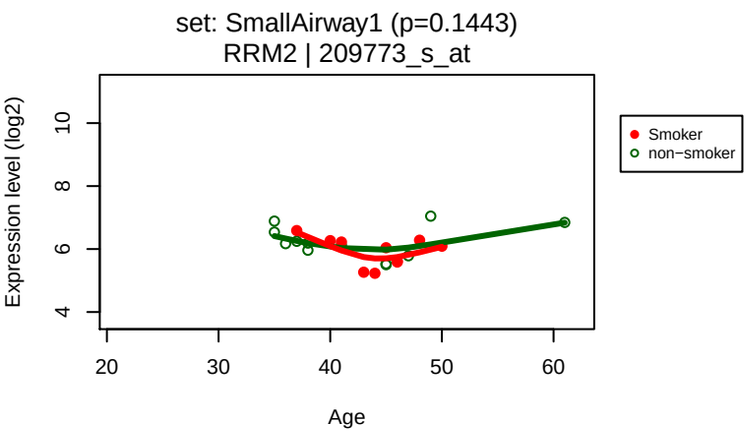
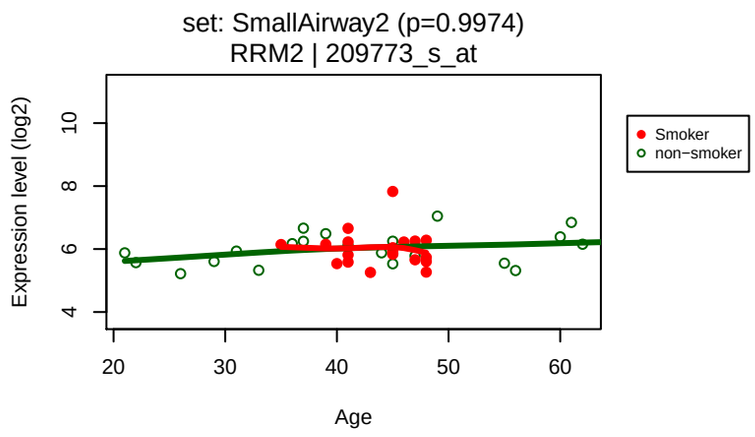
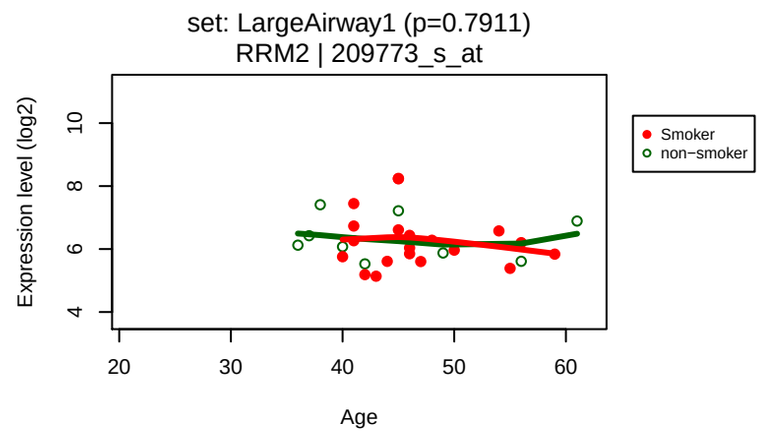

Gene expression of smokers and nonsmokers
gene: RRM2 | 209773_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1443
p=0.9974
p=0.004
p=0.7911
p=0.5134
p=0.0341
p=0.2811
p=0.401
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

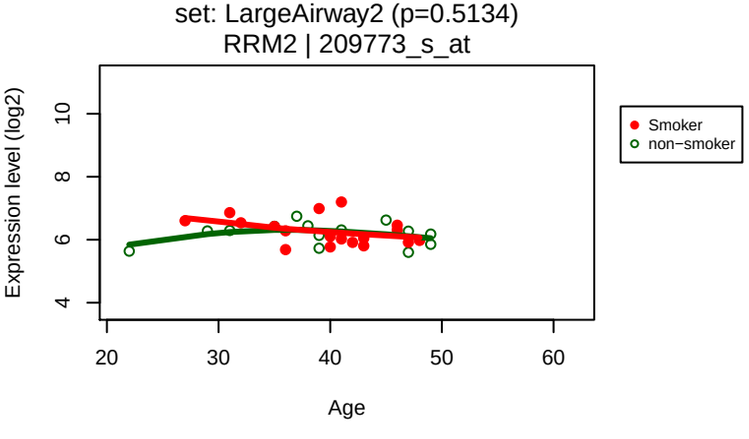

Gene expression ratio (smokers/nonsmokers)
gene: RRM2 | 209773_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1443
p=0.9974
p=0.004
p=0.7911
p=0.5134
p=0.0341
p=0.2811
p=0.401
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

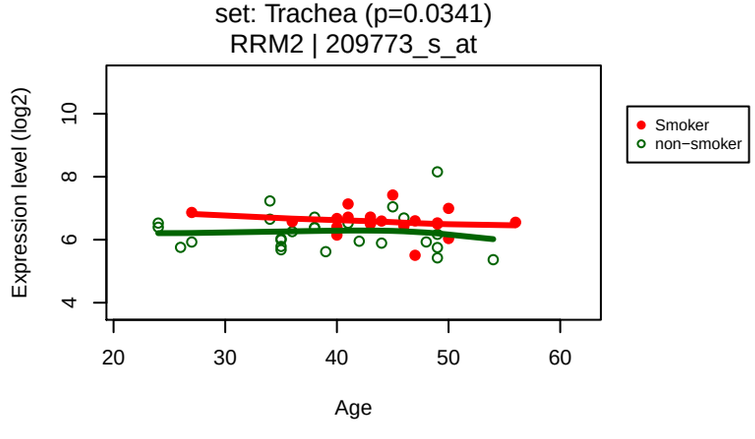

Gene expression ratio in Males
gene: RRM2 | 209773_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6267
p=0.5007
p=0.0085
p=0.319
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: RRM2 | 209773_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3562
p=0.9359
p=0.5041
p=0.8433
SmallAirway2
LargeAirway2
Trachea
Macrophage2

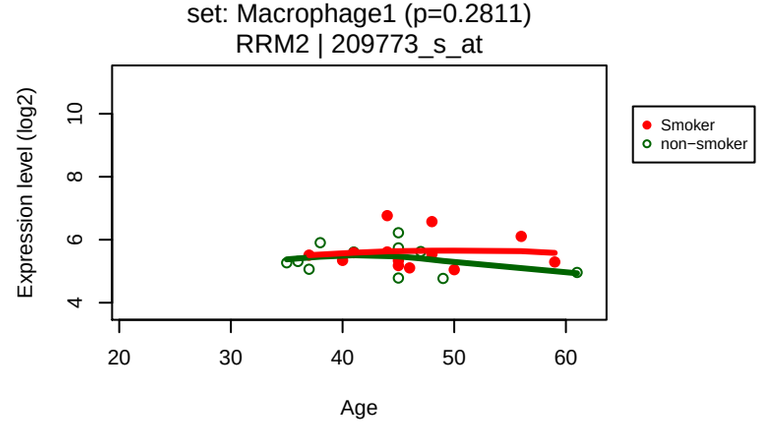

set: SmallAirway1 (p=0.1443)
RRM2 | 209773_s_at
Expression level (log2)
Age
Smoker
non−smoker

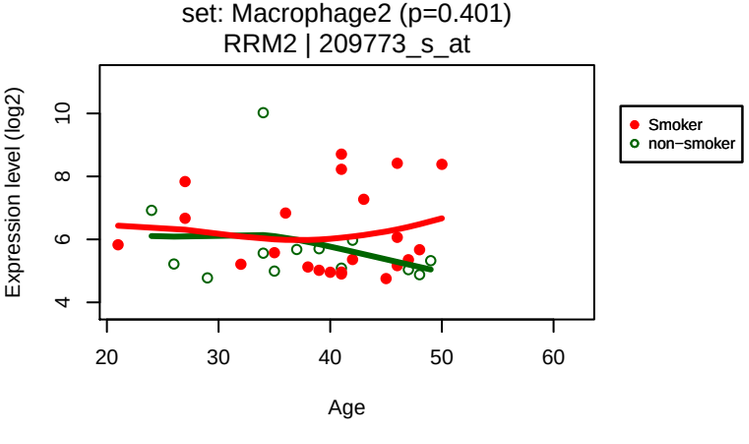

set: SmallAirway2 (p=0.9974)
RRM2 | 209773_s_at
Expression level (log2)
Age
Smoker
non−smoker

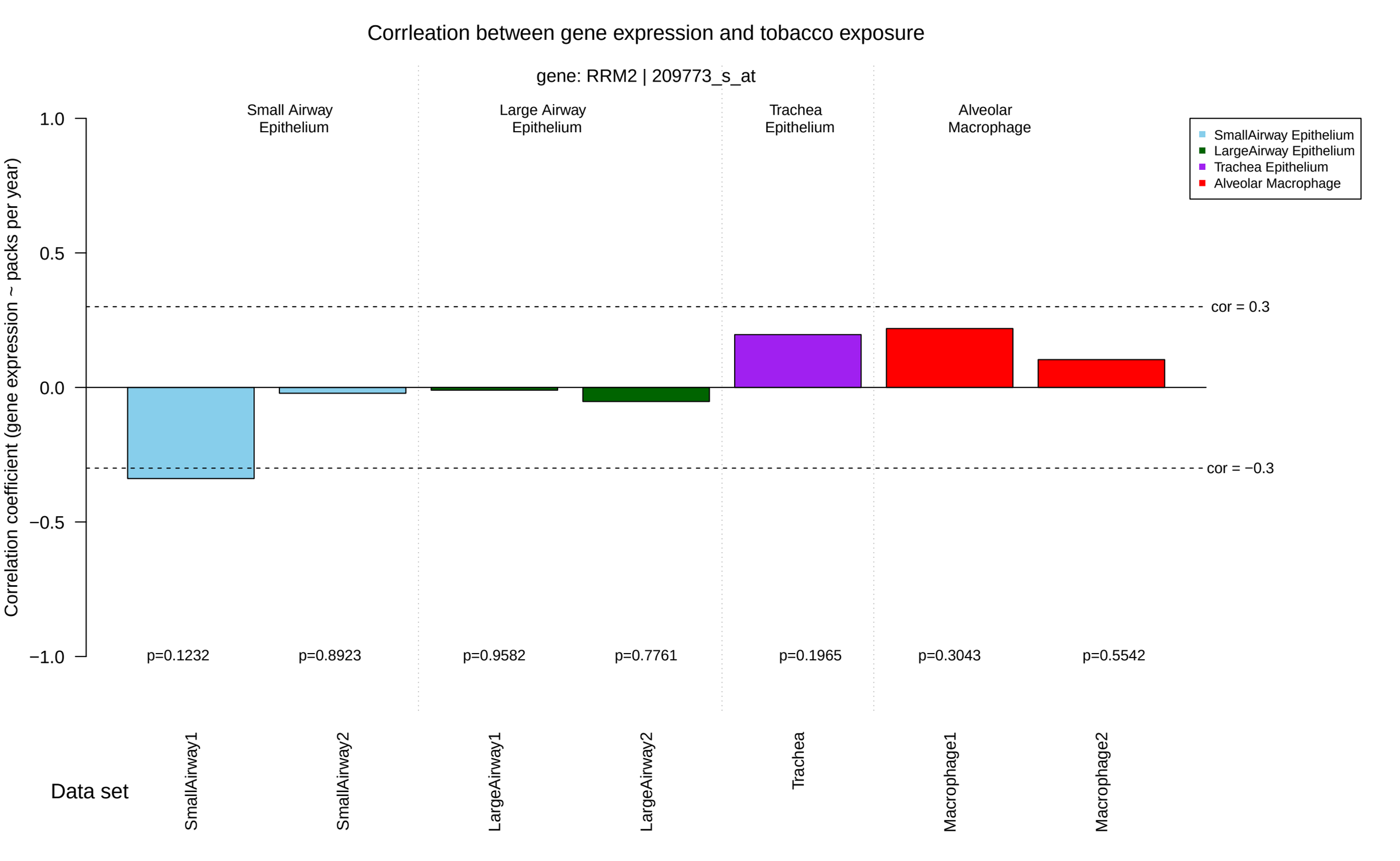

set: LargeAirway1 (p=0.7911)
RRM2 | 209773_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5134)
RRM2 | 209773_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0341)
RRM2 | 209773_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2811)
RRM2 | 209773_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.401)
RRM2 | 209773_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: RRM2 | 209773_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1232
p=0.8923
p=0.9582
p=0.7761
p=0.1965
p=0.3043
p=0.5542
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2